

MASTER BIOLOGY DIAGRAM GUIDE (CBSE 2026)

Strategic Visual Checklist & Answer Key

Analyzed by Pragya • Compiled for Thunderstudy

1. DNA Replication Fork

 VERY HIGH

[Draw Y-shaped fork: 5'-3' Continuous strand & 3'-5' Okazaki Discontinuous strand]

Key Concept

Semi-discontinuous replication in *E. coli*.

Must Label

Helicase, RNA Primer, DNA Ligase, Template strands.

2. Recombinant DNA Flow

 HIGH

[Draw Plasmid + Foreign DNA → Digestion by EcoRI → Ligation → Recombinant DNA]

Enzyme

Restriction Endonuclease (EcoRI) for cuts, Ligase for joins.

Objective

Cloning a gene of interest into a host (*E. coli*).

3. Fertilised Embryo Sac

 EXTREME

[Draw 7-celled sac: Zygote at micropyle, PEN in center, Degenerating antipodals]

Must Label

Primary Endosperm Nucleus ($3n$), Zygote ($2n$),
Degenerating Synergids.

Process

Double Fertilization Result.

4. Hershey & Chase Experiment

 VERY HIGH

[Draw Phage with ^{35}S (Coat) and ^{32}P (DNA) → Infection → Blending → Centrifugation]

Key Result

Radioactive ^{32}P detected in cells, not ^{35}S .

Conclusion

DNA is the hereditary material.

5. Pedigree Tree Analysis

 **HIGH**

[Example: Square(Male)-Circle(Female) connected by line, dark shapes = affected]

Identification

Skip gen = Recessive; In every gen = Dominant.

Tip

Look for Myotonic dystrophy (A.D.) vs Sickle cell (A.R.).



PRIORITY RANKING FOR REVISION

1. **DNA Replication Fork** (95% Probability)
2. **Embryo Sac (Plant)** (90% Probability)
3. **pBR322 Vector** (Revision needed for Section D)
4. **Hershey-Chase Chart** (Revision needed for Section E)



Smart Tip: For diagram questions, spend 70% of the time on **correct labeling** and 30% on neatness. Biology markers look for accurate labels first!

Diagram Master PDF by Pragya Thunderstudy
Source Code: 80% Diagram Accuracy Strategy